SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. **A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.**

Aim

The aim of this assignment is to analyze why a neural network achieves a low Mean Squared Error (MSE) of 0.85 during training but a much higher MSE of 5.20 during testing. The assignment also aims to identify the possible causes of this performance gap and suggest effective techniques to improve the model's ability to generalize to unseen data.

Introduction

Artificial Neural Networks (ANNs) are one of the most widely used machine learning models for solving regression and classification problems. They are inspired by the structure and functioning of the human brain, where interconnected neurons work together to process information and learn patterns from data.

In regression problems, such as predicting student examination scores, the neural network learns the relationship between input features (such as attendance, study hours, previous grades, assignments, and class participation) and the target output (student score). During training, the network adjusts its weights and biases using optimization algorithms like Gradient Descent and Backpropagation to minimize prediction errors.

The performance of a regression model is commonly evaluated using Mean Squared Error (MSE). MSE measures the average squared difference between the predicted values and the actual values. A lower MSE indicates better prediction accuracy, while a higher MSE indicates larger prediction errors.

The Mean Squared Error is calculated using the following formula:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where:

- n = Number of observations
- yi = Actual value
- ŷi = Predicted value

During model development, two different MSE values are usually observed:

- Training MSE – Error calculated using the training dataset.
- Testing MSE – Error calculated using a separate testing dataset that the model has never seen before.

Ideally, both values should be close to each other. However, in this case, the neural network has:

- Training MSE = 0.85
- Testing MSE = 5.20

This large difference indicates that the model performs extremely well on the training data but poorly on unseen testing data. Such a situation generally suggests that the model has memorized the training examples instead of learning the underlying patterns. This phenomenon is known as overfitting.

Overfitting reduces the model's ability to generalize to new data and is one of the major challenges in machine learning. It often occurs due to excessive model complexity, insufficient training data, noisy datasets, improper feature selection, or inadequate regularization.

Analyzing the reasons behind the gap between training and testing MSE is essential because it helps improve the reliability, robustness, and predictive performance of the neural network. By applying suitable corrective measures such as regularization, dropout, early stopping, data augmentation, cross-validation, and better feature engineering, the model can achieve improved generalization and more accurate predictions on unseen data.

This assignment focuses on understanding the reasons for the large difference between training and testing MSE, analyzing its impact on model performance, and suggesting suitable techniques to improve prediction accuracy and reduce overfitting.

Objectives

The objectives of this assignment are:

1. To understand the concept of Mean Squared Error (MSE).
2. To compare training and testing performance of a neural network.
3. To identify the reasons for high testing error.
4. To analyze the problem of overfitting in neural networks.
5. To study the factors affecting model generalization.
6. To suggest appropriate techniques to improve testing performance.
7. To understand the importance of regularization and validation methods.
8. To evaluate methods for building a robust neural network model.

Need for the Study

Machine learning models are expected to perform well not only on the data used for training but also on new unseen data. A model with very low training error but high testing error cannot be considered reliable. Therefore, understanding the reasons behind this difference helps developers build models that are more accurate, stable, and suitable for real-world applications such as student performance prediction, medical diagnosis, financial forecasting, and recommendation systems.

Scope of the Study

This assignment focuses on:

- Neural network regression models.
- Mean Squared Error (MSE) as the evaluation metric.
- Analysis of training and testing performance.
- Identification of overfitting issues.
- Techniques to improve model generalization.
- Practical methods for reducing prediction error.

Expected Outcome

After completing this assignment, students will be able to:

- Explain the significance of Mean Squared Error.
- Interpret differences between training and testing performance.
- Identify overfitting in neural networks.
- Recommend appropriate corrective measures.
- Build machine learning models with better generalization capability.
- Improve prediction accuracy using suitable optimization techniques.

Conclusion

A large difference between training MSE (0.85) and testing MSE (5.20) indicates that the neural network has learned the training data too specifically and fails to generalize well to new data. Understanding the causes of this problem and applying suitable corrective techniques are essential for developing reliable, accurate, and efficient machine learning models. This assignment provides a detailed analysis of the problem and explores practical solutions for improving neural network performance.

Code of Conduct:

I **D.N Shirshika (192425396***)***____________________** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: